

MUCKLE FLUGGA, United Kingdom

WMO: 030010

Lat: 60.855N

Lon: 0.886W

Elev: 53

StdP: 100.69

Time Zone: 0.00 (EUW)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.2	0.8	-4.9	2.5	2.2	-4.0	2.7	2.1	28.2	6.2	26.6	6.2	9.6	0	0.986

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.6	16.5	13.9	15.4	13.1	14.4	12.5	14.3	16.0	13.5	14.9	12.9	14.0	5.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	9.7	14.9	12.7	9.2	14.0	12.2	8.9	13.5	40.2	16.2	38.1	15.1	36.5	14.1	16.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.4	23.1	20.8	DB	-1.8	19.1	1.4	1.6	-2.8	20.2	-3.6	21.1	-4.4	22.0	-5.4	23.2	(4)
(5)			WB	-2.8	15.4	1.2	1.0	-3.7	16.2	-4.4	16.7	-5.1	17.3	-5.9	18.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	5.1	4.4	5.0	6.1	8.4	10.2	12.2	12.2	10.6	8.6	7.0	5.7	(6)	
	DBStd	3.24	1.93	1.88	2.03	1.90	2.02	1.84	1.40	1.27	1.56	1.80	1.68	1.79	(7)	
	HDD10.0	942	153	156	156	117	58	19	1	0	11	48	89	134	(8)	
	HDD18.3	3779	411	389	414	367	307	246	190	189	232	302	339	393	(9)	
	CDD10.0	204	0	0	0	0	9	23	69	69	29	5	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	9.2	12.2	10.8	11.4	8.6	7.6	6.2	6.3	6.9	9.0	10.0	10.7	11.3	(14)	
(15) Precipitation	PrecAvg	1210	137	95	118	75	56	58	59	76	115	132	144	143	(15)	
	PrecMax	1558	228	199	211	143	112	119	120	138	223	275	265	277	(16)	
	PrecMin	976	43	29	31	4	14	6	22	16	29	60	45	80	(17)	
	PrecStd	142	48	42	39	30	25	25	23	31	42	46	45	47	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	9.8	10.2	11.7	17.5	17.7	18.1	17.8	15.6	12.4	11.0	10.0	(19)	
		MCWB	8.4	8.3	8.4	9.2	13.8	14.0	14.7	15.6	13.3	10.8	10.3	9.1	(20)	
	2%	DB	9.1	8.7	9.1	10.2	15.1	15.5	16.5	15.7	14.1	11.7	10.2	9.2	(21)	
		MCWB	8.3	7.6	7.8	8.2	12.0	12.6	13.8	13.9	12.3	10.4	9.5	8.3	(22)	
	5%	DB	8.4	7.7	8.3	9.3	12.2	14.0	15.3	14.8	13.3	11.2	9.8	8.8	(23)	
		MCWB	7.5	6.5	7.2	7.8	10.0	11.6	13.2	13.1	11.8	10.2	9.0	7.8	(24)	
	10%	DB	7.7	7.1	7.7	8.6	10.8	12.7	14.3	14.0	12.6	10.8	9.4	8.3	(25)	
		MCWB	6.6	6.0	6.5	7.2	8.9	10.9	12.5	12.5	11.3	9.8	8.4	7.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	8.6	8.8	9.6	14.1	14.2	15.2	15.8	13.7	11.4	10.5	9.4	(27)	
		MCDB	9.3	9.2	9.6	11.2	17.6	17.2	17.2	17.0	14.7	12.0	10.7	9.6	(28)	
	2%	WB	8.4	7.9	8.1	8.7	12.2	12.8	14.3	14.2	12.8	10.9	9.8	8.7	(29)	
		MCDB	8.9	8.6	8.8	9.6	14.7	15.0	16.0	15.3	13.8	11.3	10.1	9.2	(30)	
	5%	WB	7.7	6.9	7.5	8.2	10.3	12.0	13.5	13.4	12.1	10.5	9.2	8.0	(31)	
		MCDB	8.3	7.5	8.2	9.0	11.8	13.5	14.9	14.3	12.9	11.1	9.7	8.7	(32)	
	10%	WB	6.8	6.2	6.7	7.5	9.2	11.2	12.7	12.9	11.6	10.0	8.7	7.4	(33)	
		MCDB	7.6	7.0	7.6	8.5	10.3	12.5	13.9	13.7	12.4	10.8	9.4	8.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	2.2	2.4	2.6	2.9	3.2	3.0	3.0	2.6	2.5	2.0	2.0	2.4	(35)	
		MCDBR	3.0	3.1	3.1	3.9	5.4	5.2	4.7	3.7	3.3	2.2	2.4	2.9	(36)	
	5% WB	MCWBR	3.6	3.4	2.9	3.1	3.8	3.4	3.0	2.6	3.0	2.3	2.7	3.6	(37)	
		MCWBR	2.9	3.2	2.9	3.6	4.9	4.5	4.3	3.3	2.9	2.2	2.4	3.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.330	0.352	0.366	0.379	0.370	0.363	0.377	0.364	0.361	0.344	0.321	0.398	(40)	
		taud	2.071	2.196	2.276	2.313	2.404	2.463	2.443	2.478	2.441	2.317	2.126	2.358	(41)	
	Ebn at Noon	Ebn at Noon	362	605	745	813	854	867	843	826	755	617	371	232	(42)	
		Edh at Noon	45	72	91	105	104	99	99	89	78	63	41	30	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.21	0.70	1.77	3.34	4.67	4.83	4.35	3.55	2.14	1.06	0.31	0.13	(44)	
	RadStd	RadStd	0.02	0.08	0.14	0.20	0.33	0.53	0.38	0.34	0.20	0.11	0.03	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)		N/A	+0.65	+0.62	N/A	N/A	N/A	-102	-134	N/A	N/A

Nomenclature: See separate page